

Covariant Differentiation from the AASC Kernel

Standing-Preserving Transport, Identity Principal Symbol, and Levi-Civita Closure

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Abstract

AASC is a role-constraint formalism for the necessity conditions of non-degenerate determinate construction. It is not an interpretive overlay placed on independently complete physics. The canonical kernel theorem establishes, by fixed-domain exhaustion, that determinate same-domain use requires Admissibility, Standing, Reference, and Irreversibility/Fixation. This paper asks a narrower question: once a governed domain occupies the covariant-differentiation target, what smooth local first-order realization can faithfully represent the standing-preserving transport-comparison role?

The proof separates role necessity, realization existence, and representative classification. First, target occupation forces a pre-differential transport-comparison role: standing-bearing carrier content cannot be compared across admitted continuation by raw representative comparison, static transition bookkeeping, privileged representatives, witnessless assignment, same-act repair, or undeclared scope enrichment. Second, smooth realization is a branch condition: the kernel does not numerically select a connection, a gauge group, a global bundle topology, or a dynamical action. Third, in the standard finite linear carrier branch, an admissible smooth realization is not an arbitrary first-order operator. It is the canonical infinitesimalization of finite standing-preserving linear transport. Identity, restriction, reversal, weak composition, smooth first-jet dependence, lawful-redescription equivariance, and preservation of the already fixed finite linear carrier operations force additivity, direction linearity, and the identity principal symbol

$$\sigma_{\nabla}(\xi) = \xi \otimes \text{id}_E .$$

Consequently the difference from the raw representative derivative is C^∞ -linear in both the section and direction arguments, and locally

$$\nabla = d + A, \quad A \in \Omega^1(\text{End } E).$$

Gauge, tangent, and tensor covariant derivatives are typed specializations. Metricity remains downstream from governance. In the minimal first-order natural metric branch, metric comparison preservation gives $\nabla g = 0$, and a natural difference-tensor theorem excludes every nonzero first-order natural metric-compatible contorsion, whether primitive or derived. Hence $\nabla = \nabla^{LC}$ in that branch. Torsionful, metric-affine, higher-jet, nonlocal, and modified carrier theories are not denied; they are declared richer-scope or higher-response branches. The novelty is not the familiar local formula. It is the dependency theorem showing that connection type is the forced smooth first-jet realization of kernel-forced standing-preserving transport once the covariant-differentiation target, standard finite linear carrier branch, and admissible smooth transport realization are occupied.

Keywords: AASC; covariant derivative; connection; standing-preserving transport; first jet; principal symbol; fixed-domain exhaustion; descent; quotient; gauge covariance; Levi-Civita connection; metric compatibility; natural operations; contorsion; determinate objecthood.

1 Introduction

The covariant derivative is usually introduced after a smooth manifold, vector bundle, transition functions, and a connection have already been granted. That route is efficient, but it hides the dependency problem. Why must standing-bearing carrier content be compared by transport rather than by raw representatives, and why must the smooth first-order realization of that transport have connection type?

AASC answers the prior question. Its kernel is not generated by metric geometry. It identifies the governance roles already required by determinate same-domain construction: fixed target reference, consequence-bearing standing, admissible transformation, and act-time fixation. Smooth manifolds, bundles, gauge groups, metrics, numerical connection fields, curvature, and field equations are downstream realizations or specializations.

The dependency established here is not

$$\text{standard differential geometry} \Rightarrow \text{AASC interpretation.}$$

It is the typed chain

$$\text{DetUse}(D) \Rightarrow K_D \Rightarrow \mathcal{Q}_D,$$

$$\text{CovDiffTarget}(D) \Rightarrow \text{TC}_D,$$

and, in the standard finite linear branch with an admissible smooth first-jet realization,

$$\text{TC}_D + \text{Real}_1^{\text{tr}}(D) + \text{StdFinLin}(D) \Rightarrow \nabla = d + A.$$

The second arrow is a role-necessity theorem. The third is a realization-classification theorem. Neither claims that the kernel by itself selects a particular numerical connection, a global topology, a coupling constant, a metric signature, or a dynamical law.

Raw differentiation fails exactly where lawful redescription has nontrivial first-order content. If a local gauge representative changes by $\psi' = g\psi$, then

$$\partial_\mu \psi' = g \partial_\mu \psi + (\partial_\mu g) \psi.$$

The second term differentiates the redescription rather than the standing-bearing carrier content. Similarly, $\partial_\mu V^\nu$ fails to transform tensorially under a general coordinate change. Conventional geometry repairs these failures by introducing a connection. The present theorem locates the prior dependency: once a carrier must retain standing through admitted continuation and lawful local redescription, no weaker same-scope comparison route survives; and once that role is realized as smooth finite linear transport to first order, the infinitesimal operator is a connection.

A proof that treats the smooth operator directly is incomplete unless it fixes two mathematical points. First, it must force the identity principal symbol. Operators such as $c\nabla$ with $c \neq 1$ are local, smooth, covariant, and linear, but they do not have the correct scalar first-jet normalization. Second, it must exclude nonlinear first-order operators made from already declared metric data. This paper therefore makes finite transport primary. The connection identities are derived by differentiating lawful finite transport, not by declaring a generic first-order operator minimal.

The paper proceeds as follows. Sections 2-5 fix the proof class, kernel dependency, scope ledger, descent interface, and standard finite linear branch. Sections 6-7 derive the pre-differential transport-comparison role and finite route closure. Sections 8-13 establish the smooth transport realization, identity principal symbol, full connection identities, and local normal form. Section 14 integrates same-scope operator exhaustion to classify nonconnection operator-family deviations without substituting role language for differential analysis. Sections 15-17 give the gauge, tangent, tensor, and metric branches, including the natural Levi-Civita closure. Sections 18-21 state curvature/dynamics consequences, the main theorem, richer-scope boundaries, and the exact refutation burden. Appendices record the finite transport antecedent, the adversarial regression suite, and conventional normalization.

2 Proof class, exact claim, and imported dependencies

2.1 Proof-class lock

AASC is a constraint formalism. Its native proof form is fixed-domain exhaustion and impossibility closure. The proof does not construct the kernel from a lower layer that is already using the same governance work. It asks whether the fixed target can survive removal, weakening, deferral, externalization, repair, or replacement of a role. A route fails when it loses the target, changes domain or act identity, imports substitute governance, collapses to bookkeeping, or reproduces governance-equivalent work.

This is role-minimality under a fixed target. It is not the claim that four uninterpreted predicates are syntactically independent in every formal language. A countermodel that removes Reference by also removing fixed target identity, or removes Fixation by allowing a later act to replace the old act, has changed the target and is not a same-regime counterexample.

Proof-Class Lock 2.1. A criticism of a load-bearing step has standing only if it does at least one of the following:

1. denies that the stated target is instantiated;
2. breaks an imported upstream theorem on its own comparison class;
3. identifies an invalid dependency or specialization step;
4. supplies a faithful same-regime survivor preserving the frozen target, carrier, descent interface, continuation profile, standing-bearing use, operator grammar, first-jet normalization, and fixation regime; or
5. identifies a mathematical defect in the local realization theorem.

Preference for a metric-first derivation, a Lovelock-style uniqueness proof, or a constructive physical dynamics is not a correctness objection unless it identifies one of those failures.

2.2 The exact theorem claim

Three propositions are separated throughout the paper.

Role necessity. If the covariant-differentiation target is occupied, standing-preserving cross-local comparison must factor through a pre-differential transport-comparison role TC_D .

Realization existence. Whether a smooth first-order realization exists depends on the realization branch. The kernel does not produce a numerical connection. In the standard paracompact smooth vector-bundle branch, the conventional existence theorem for connections supplies examples.

Representative classification. Every admissible smooth first-jet transport realization in the standard finite linear carrier branch has local connection form $d + A$.

The paper proves the first and third claims. It records conventional existence only to identify the familiar mathematical branch in which the forced role is realized.

2.3 Imported and new results

The following results are imported at their stated type.

- The four-role fixed-domain kernel theorem and anti-weakening burden [1].
- Same-scope operator closure, graph rigidity, family exhaustion, no internalized import, no post-hoc repair, and no deferred admissibility [2].

- Fixed-domain comparison class, scope fixation, factorization discipline, and fail-closed typing [3].
- Standing quotient, skin non-authority, and anchor/tensor/skin role decomposition [4].
- Finite standing-preserving transport geometry and the pre-differential connection role [5].
- Metricity as licensed downstream description and no metric backflow [6].
- The metric projection and first-response scope discipline used to separate minimal metric grammar from richer-scope modified geometry [7].

The new contribution is the differential bridge: the exact covariant ledger including operator grammar and first-jet normalization, smooth transport realization, carrier-morphism theorem, canonical infinitesimalization, identity principal-symbol theorem, full connection identities, and the natural first-order Levi-Civita closure.

3 Kernel dependency and cost of denial

3.1 Neutral target adequacy

The kernel is not rederived here from definitions that already name its roles. The upstream theorem begins with a neutral target class.

Definition 3.1 - Determinate same-domain use. $\text{DetUse}(D)$ holds when a regime D has:

1. **non-degeneracy:** at least one distinction changes a consequence-bearing constructional result;
2. **target determinacy:** the object, carrier, act, or endpoint under evaluation is fixed as the same target across the claimed use;
3. **transition evaluability:** admitted and non-admitted uses are distinguishable on the fixed domain;
4. **act-time finality:** an evaluated act cannot be retrospectively converted into the same admissible act by later repair; and
5. **same-regime fidelity:** identity-preserving redescription preserves target identity, act identity, standing-bearing status, continuation profile, and reportability.

Imported Theorem 3.2 - Kernel Necessity and Anti-Weakening. If $\text{DetUse}(D)$ holds, then the fixed-domain kernel

$$K_D = (\text{Adm}_D, \text{Stand}_D, \text{Ref}_D, \text{Fix}_D)$$

is instantiated up to governance equivalence. Every strict same-domain weakening loses the target, imports substitute governance, changes act or domain, collapses to bookkeeping, or reproduces governance-equivalent work.

The theorem is role-minimal. Each role is necessary for a distinct target function. It is not arbitrary predicate independence.

3.2 Cost of role denial

The derivative theorem uses every kernel role.

Reference denial. Without Reference there is no fixed carrier whose variation is being compared:

$$\neg \text{Ref}_D \Rightarrow \neg \text{FixedCarrierTarget}(D).$$

Standing denial. Without Standing, comparison outputs carry no consequence-bearing or reuse-bearing force:

$$\neg \text{Stand}_D \Rightarrow \neg \text{ConsequenceBearingComparison}(D).$$

Admissibility denial. Without Admissibility, licensed redescription, arbitrary replacement, hidden selector import, same-act repair, and scope enrichment are not distinguishable:

$$\neg \text{Adm}_D \Rightarrow \neg \text{FixedComparisonBoundary}(D).$$

Fixation denial. Without Fixation, an eliminated route can be reopened, a witnessless assignment can be retrospectively certified, and an enriched carrier can be relabeled as though it had remained inside the original act:

$$\neg \text{Fix}_D \Rightarrow \neg \text{StableRouteClosure}(D).$$

3.3 Fixation closes the exhaustion

Theorem 3.3 - Exhaustion Closure Requires Fixation. Let a fixed comparison audit assign terminal dispositions to candidate routes. If Fixation is absent, the audit does not yield a closed endpoint classification.

Proof. Route exhaustion requires that representative-sensitive comparison remain excluded, selector import remain classified as import, witnessless comparison remain non-admitted, and scope enrichment remain a new scope. Without Fixation, any of these classifications may be reopened within the same act: a later selector may be treated as if it had been present from the start, a later witness may retroactively certify an earlier pathless comparison, or an enriched carrier may be reidentified as the original carrier. The same-act status of the candidate is therefore not fixed. With Fixation, later support may create a new act or richer regime but cannot repair the status of the evaluated act. $\square$

Fixation is therefore load-bearing in the covariant-differentiation proof. It is not thermodynamic irreversibility and not a claim that physical trajectories are dynamically irreversible.

4 Frozen scope, operator grammar, and descent

4.1 The covariant realization ledger

A same-scope theorem cannot classify scope after seeing which candidate survives. The comparison background is frozen first.

Definition 4.1 - Covariant realization ledger. The frozen covariant ledger is

$$\mathfrak{S}_D^\nabla = (T_D, C_D, \mathcal{P}_D, \mathcal{Q}_D, \text{Adm}_D, \mathcal{L}_D, \mathcal{R}_D, \mathcal{F}_D, \mathcal{O}_D, J_D),$$

where:

- T_D is the fixed first-order comparison target;
- C_D is the standing-bearing carrier class;
- $\mathcal{P}_D$ is the local presentation category or groupoid;
- $\mathcal{Q}_D$ is the standing-bearing descent interface;
- Adm_D is the admissibility standard;
- $\mathcal{L}_D$ is the redescription and continuation ledger;
- $\mathcal{R}_D$ is the transport-comparison role to be realized;
- $\mathcal{F}_D$ is the fixation and terminal-classification regime;
- $\mathcal{O}_D$ is the declared operator family and carrier morphism graph; and
- J_D is the first-jet normalization anchor.

In the standard finite linear branch, $\mathcal{O}_D$ contains the zero, addition, scalar-multiplication, restriction, and local linear morphism structure of the carrier. The first-jet anchor records that base scalars vary according to the tangent action

$$J_D(X, f) = X[f].$$

Definition 4.2 - Same-scope realization. Two candidates are background-same-scope when they are evaluated against the same $\mathfrak{S}_D^\nabla$. A background-same-scope candidate is standing-preserving when it also preserves target identity, carrier role, descent identity, admitted continuation, first-jet normalization, operator grammar, and terminal status without imported data.

Definition 4.3 - Scope enrichment. $\mathfrak{S}_D^\nabla \prec \mathfrak{S}_D^{\nabla'}$ when the latter adds or changes an independent carrier, selector, operator family, first-jet normalization, connection tensor, source, boundary datum, nonlocal rule, comparison rule, differential order, or standing-bearing occupant absent from the former.

Thus extra load is not restricted to new tensor fields. A candidate that preserves all tensors but changes the operator graph, principal symbol, carrier morphism class, or differential order changes scope.

4.2 Anchor, tensor, and skin typing

The proof uses the anchor/tensor/skin decomposition of [4]. Anchors are preconditions of admissibility and target fixation. Tensors are quotient-level standing-bearing invariants. Skin is presentation surplus. A proposed fourth same-scope role either changes an anchor, changes tensor content, or changes neither and is skin.

For the present theorem, the carrier operations and the first-jet scalar action are anchors. A connection difference A is tensor-level representative content when it realizes the already-forced transport role. A local frame, coordinate chart, or gauge representative is skin unless it changes the descent or transport ledger.

4.3 Descent rather than a naive orbit set

A set quotient is too weak for local differential geometry and may be badly behaved in gauge problems. The theorem therefore uses a faithful descent interface.

Definition 4.4 - Faithful standing descent. A faithful standing descent consists of a presentation category or groupoid $\mathcal{P}_D$, a standing-bearing target $\mathcal{Q}_D$, and a functor or quotient map

$$\Pi_D : \mathcal{P}_D \longrightarrow \mathcal{Q}_D$$

such that:

1. presentation-equivalent objects descend to the same object or to canonically specified isomorphic objects;
2. descent commutes with admissible restriction;
3. continuation induces a typed comparison problem on descended content; and
4. every admissible presentation-invariant standing-bearing comparison factors through Π_D .

In ordinary quotient branches the equality notation $\Pi_D(x) = \Pi_D(y)$ is literal. In groupoid or associated-bundle branches it means a specified isomorphism in the descent object. The load-bearing theorem $\nabla = d + A$ is proved only in the smooth finite-rank vector-bundle branch, represented by a local presentation groupoid and an arrow congruence compatible with source, target, identity, inverse, restriction, and partial composition. More general sheaf-like or stack-like descent interfaces are possible extensions; no stack-level theorem is claimed here.

5 The standard finite linear carrier branch

The familiar formula $d + A$ is not asserted in every determinate domain. It belongs to the standard finite linear carrier branch.

Definition 5.1 - Standard finite linear carrier branch. $\text{StdFinLin}(D)$ holds when the covariant-differentiation target is realized over a smooth base M by a fixed finite-rank locally free C_M^∞ -module of sections, equivalently by a smooth finite-rank vector bundle $E \rightarrow M$, with:

1. fixed local rank r ;
2. local addition, zero, and scalar multiplication by base functions;
3. compatible restriction maps;
4. smooth $GL(r)$ -valued transition functions;
5. identity and cocycle laws on overlaps;
6. quotient-preserving redescription action; and
7. carrier-role preservation under admitted continuation.

This is a branch definition, not a kernel theorem. The kernel forces the transport-comparison role when the covariant-differentiation target is occupied. The standard branch supplies the conventional smooth finite linear carrier on which that role can be realized by transport maps and a connection.

The standard finite linear branch does not assert that every determinate domain is linear. It names the branch in which the covariant-differentiation target is carried by finite-dimensional local linear carrier structure. Within that branch, $0, +, \cdot$, restriction, smooth local transition, and quotient-preserving overlap maps are part of the fixed carrier graph. The theorem below does not derive linear carrier structure from the bare kernel. It proves that once this branch is occupied, a same-scope transport realization cannot alter that carrier graph without becoming a different carrier category, an operator-family enrichment, or a declared scope change.

Proposition 5.2 - Bundle reconstruction in the standard branch. A fixed finite-rank locally free C_M^∞ -module with smooth $GL(r)$ cocycles and compatible restrictions is represented by a smooth vector bundle $E \rightarrow M$, and its local sections are the carrier representatives in $\text{StdFinLin}(D)$.

Proof. This is the standard equivalence between finite-rank vector bundles and locally free finite-rank sheaves of smooth sections with smooth transition cocycles. The AASC contribution is not to reprove that equivalence, but to identify when the branch is the faithful smooth carrier presentation of the standing-bearing role. $\square$

5.1 Assumption/derivation ledger

The proof uses the following fixed/derived separation.

- **Kernel layer.** The fixed target is non-degenerate determinate use. The imported kernel theorem yields $K_D = (\text{Adm}, \text{Stand}, \text{Ref}, \text{Fix})$.
- **Covariant target layer.** The fixed target is first-order local comparison. This yields the continuation, transportable-carrier, and smooth first-order burdens.
- **Standard finite linear branch.** The fixed branch supplies finite-rank local linear carrier structure. Same-scope transport must preserve that carrier graph.
- **Finite transport layer.** The fixed problem is admitted continuation and quotient-facing comparison. The finite route theorem yields TC_D as the only same-scope finite survivor.
- **Smooth realization layer.** The fixed realization data are finite transport maps with identity, reversal, restriction, composition, redescription equivariance, carrier-graph preservation, and first-jet locality. The derived object is the canonical infinitesimalization.
- **Operator layer.** The fixed graph is the first-jet transport-realization class with $J_D(X, f) = X[f]$. The derived results are identity principal symbol, section additivity, and direction-linearity.
- **Local presentation layer.** A local trivialization is fixed only as presentation. The derived local normal form is $\$ \text{abla} = d + A \$$ with $A \in \Omega^1(\text{End } E)$.
- **Metric branch.** The fixed branch is TM , g , and $\text{MinMetricNat}_1(M, g)$. The derived result is Levi-Civita closure.

6 Target occupation and the pre-differential transport role

6.1 Target partition

The kernel does not imply that every determinate domain has a covariant derivative.

Definition 6.1 - Target partition. A governed domain may occupy one of the following target levels.

1. **Kernel-only target:** determinate governed use without cross-local transport.
2. **Finite transport target:** standing-bearing content is compared across admitted continuation, but no smooth first-order local refinement is required.
3. **Covariant-differentiation target:** standing-bearing carrier content must be compared to first order under lawful local redescription.

Definition 6.2 - Covariant-differentiation target. $\text{CovDiffTarget}(D)$ holds when D contains standing-bearing carrier content for which a same-scope first-order comparison is required under lawful local redescription.

The definition fixes the mathematical target. Denying first-order local variation

does not supply a rival covariant derivative. It retypes the problem as finite transport or generalized comparison.

Lemma 6.3 - Target-forced refinement.

$$\text{CovDiffTarget}(D) \Rightarrow \text{Cont}_D + \text{TransportableCarrier}(D) + \text{Smooth}_1(D).$$

Proof. A covariant derivative compares variation of the same standing-bearing carrier along a local displacement. Without admitted continuation there is no same-domain displacement. Without transportable carrier content there is no retained carrier to compare. Without first-order local refinement the target is not covariant differentiation but finite or discrete transport. $\square$

6.2 The transport-comparison role

Definition 6.4 - Pre-differential transport-comparison role. TC_D is the role assigning to admitted continuation arrows the standing-preserving comparison class required to identify transported carrier content without selecting a privileged representative. It is not a connection one-form, a list of coefficients, or a smooth derivative.

Imported Theorem 6.5 - Finite Transport-Comparison Necessity. Let D instantiate the kernel, faithful standing descent, admitted continuation, local presentation freedom, and the fixed finite transport comparison class. Every same-scope standing-preserving comparison of carrier content across admitted continuation factors through TC_D .

The proof is the finite route exhaustion: direct representative comparison is skin-sensitive; static transition bookkeeping lacks transported content; global preferred representatives import selectors; pathless assignment lacks a licensed continuation witness; post-hoc repair violates Fixation; and enriched tensor or carrier structure changes scope.

7 Finite route closure and operator-family closure

7.1 Coverage classifier for finite comparison

The finite comparison theorem uses a coverage classifier rather than a mutually exclusive taxonomy. A candidate comparison over a continuation arrow is assigned one or more dispositions:

$$\{\text{rep}, \text{book}, \text{sel}, \text{nowit}, \text{med}, \text{enrich}, \text{repair}, \text{skin}, \text{fail}\}.$$

Here med is the terminal surviving case: quotient-preserving mediated continuation realizing TC_D . Other active dispositions are terminal failures or scope changes. The classifier need not be exclusive. A candidate may be both selector-dependent and scope-enriched; either terminal disposition blocks same-scope survival.

Theorem 7.1 - No Seventh Finite Route. In the fixed finite comparison class, every admitted cross-local comparison candidate receives at least one terminal disposition. The only same-scope standing-preserving survivor is mediated quotient-preserving continuation, i.e. TC_D .

Proof. The candidate dependency signature contains representatives, transition bookkeeping, selectors, continuation witnesses, mediator data, enrichment data, and repair status. If no mediator is present, the candidate is representative-sensitive, bookkeeping-only, selector-dependent, witnessless, repair-dependent, enriched, skin, or fail-closed. If a mediator satisfying quotient-domain, continuation, carrier-role preservation, no-selector, and no-extra-load conditions is present, it is precisely TC_D . Fixation prevents reclassification after failure. $\square$

7.2 Operator-family closure is a second stage

The finite theorem forces a transport role. It does not by itself classify every differential operator. A second closure is required after the smooth realization ledger is fixed.

Definition 7.2 - Realization-family universe. $\mathfrak{R}_D^1$ is the class of smooth local first-jet realizations of TC_D on the fixed standard finite linear carrier that preserve the carrier morphism graph and the first-jet normalization anchor.

Imported Theorem 7.3 - Same-Scope Operator Closure. On a fixed standing-bearing carrier, any purported same-scope operator-family extension is either bookkeeping-equivalent, standing-fixed, a conservative definitional extension, a nonconservative family change, imported data, carrier collapse, domain enlargement, evaluative completion, or inadmissible repair. Composition, iteration, diagonalization, representation change, and foundation swap do not create a new same-scope admissible family.

Corollary 7.4 - Operator deviation classification. A differential candidate outside $\mathfrak{R}_D^1$ is classified as one of:

1. lawful redescription of the same realization;
2. representative-level bookkeeping;
3. imported normalization or selector;
4. primitive operator-graph change;
5. higher-jet or nonlocal enrichment;
6. nonlinear carrier-category enrichment; or
7. explicit scope change.

This corollary is not a substitute for differential analysis. It is applied only after Sections 8-12 prove which operators arise from admissible finite linear transport.

8 Smooth transport realization

The realization object is finite transport first and a differential operator second.

Definition 8.1 - Smooth transport realization. In the standard branch, an admissible smooth realization of $\mathbf{TC}_D$ consists of transport maps

$$P_\gamma : E_{\gamma(0)} \longrightarrow E_{\gamma(1)}$$

for admitted sufficiently regular path germs or continuation germs γ , satisfying:

1. identity: $P_{\text{const}_p} = \text{id}_{E_p}$;
2. weak composition: $P_{\gamma_2 \star \gamma_1} = P_{\gamma_2} \circ P_{\gamma_1}$ when the concatenation is admitted;
3. reversal: $P_{\gamma^{-1}} = P_\gamma^{-1}$;
4. restriction compatibility on path segments;
5. smooth dependence on path and initial carrier data;
6. locality;
7. lawful-redescription equivariance;
8. dependence to first order only on the first jet of the path;
9. preservation of the fixed finite linear carrier operations; and
10. local tangent-germ closure: admitted first jets at p are closed under sufficiently small scalar rescaling and addition, as required for the full standard first-order covariant-differentiation target.

The local tangent-germ closure clause is not a global reachability claim. It is the local tangent-domain closure required by a standard first-order differential target. If a domain lacks this local tangent-germ closure, it may support finite transport or constrained directional comparison, but not the full standard covariant-differentiation target.

We use the convention that $\gamma_2 \star \gamma_1$ means first traverse γ_1 , then γ_2 .

Equivalently, in the branch where thin homotopies are identified, the realization is a smooth transport functor

$$P : \Pi_1^{\text{thin}}(M) \longrightarrow \mathbf{CarIso}_D$$

with $\mathbf{CarIso}_D$ the category of carrier-role-preserving isomorphisms. The next theorem proves that, in $\mathbf{StdFinLin}(D)$, these morphisms are linear isomorphisms.

Remark 8.2. The identity, reversal, and composition laws are finite transport laws. The identity principal symbol is not assumed here. It is derived in Section 11 by differentiating the finite transport maps.

9 Carrier morphisms and finite linearity

Theorem 9.1 - Same-Scope Carrier-Morphism Preservation Theorem. In $\text{StdFinLin}(D)$, every admissible transport map P_γ preserves the linear carrier operations:

$$P_\gamma(0) = 0,$$

$$P_\gamma(u + v) = P_\gamma(u) + P_\gamma(v),$$

$$P_\gamma(\lambda u) = \lambda P_\gamma(u).$$

Hence $P_\gamma \in \text{Iso}_{\mathbb{F}}(E_{\gamma(0)}, E_{\gamma(1)})$.

Proof. In the standard finite linear branch, zero, addition, and scalar multiplication are standing-bearing carrier operations already fixed by the carrier graph. The theorem does not manufacture those operations; it proves that a same-scope transport realization cannot change them. A map that fails to preserve them does not transport the same linear carrier; it changes the carrier graph. The operator-exhaustion theorem then classifies the possibilities. If the altered graph is only notational, it is bookkeeping-equivalent. If it changes how carrier combinations stand, it is a nonconservative operator-family change. If it uses an undeclared weighting, selector, or nonlinear rule, it imports data. If nonlinear transport is declared as the morphism class, the theory has moved to a nonlinear carrier category. None is a same-scope realization of the fixed standard finite linear carrier. Therefore admitted same-scope transport preserves the carrier operations and is fiberwise linear. Since reversal gives an inverse transport, it is a linear isomorphism. $\square$

Corollary 9.2 - Section-level additivity under transport. For sections s, t along a path segment,

$$P_t^{-1}(s(\gamma(t)) + t(\gamma(t))) = P_t^{-1}s(\gamma(t)) + P_t^{-1}t(\gamma(t)).$$

This identity is finite. The differential additivity theorem is obtained by differentiating it.

10 Canonical infinitesimalization

Definition 10.1 - Canonical infinitesimalization. Let $s \in \Gamma(E)$ and $X_p \in T_p M$. Choose an admitted smooth path germ γ with $\gamma(0) = p$ and $\dot{\gamma}(0) = X_p$. Write $P_t = P_{\gamma|_{[0,t]}}$. The infinitesimal realization of finite transport is

$$(\nabla_X s)_p = \left. \frac{d}{dt} \right|_{t=0} P_t^{-1} s(\gamma(t)).$$

Theorem 10.2 - Well-defined Infinitesimalization. The operator ∇ defined above is independent of the chosen path germ with the same first jet, independent of local frame, compatible with restriction, and determined by X_p , $s(p)$, and the first jet of s at p .

Proof. First-jet path dependence in the realization states that two admitted path germs with the same initial point and velocity determine the same first-order transport coefficient. Restriction compatibility gives the same derivative after shrinking the domain. Lawful-redescription equivariance sends both P_t and the local representative of s through the same carrier isomorphism, so the derivative is frame-independent. Smooth dependence on path and initial carrier data gives existence of the derivative. Since the expression differentiates $s(\gamma(t))$ and the first-order transport correction at $t = 0$, no higher jet of s or γ enters. $\square$

Proposition 10.3 - Local representative form before normalization. Relative to a local frame, finite transport has an expansion

$$P_t^{-1} = I + tA_p(X_p) + o(t),$$

where $A_p(X_p) \in \text{End}(E_p)$ is smooth in p and depends only on the first jet X_p of the path.

This is not yet the connection formula. It is the first-order expansion of finite linear transport.

11 Identity principal symbol

Theorem 11.1 - Identity Principal-Symbol Theorem. The canonical infinitesimalization of admissible smooth finite linear transport satisfies

$$\nabla_X(fs) = X[f]s + f\nabla_X s.$$

Equivalently,

$$\sigma_\nabla(\xi) = \xi \otimes \text{id}_E.$$

Proof. Let $p = \gamma(0)$ and $\dot{\gamma}(0) = X_p$. Since P_t is fiberwise linear and acts only on the carrier fiber,

$$P_t^{-1}(f(\gamma(t))s(\gamma(t))) = f(\gamma(t))P_t^{-1}s(\gamma(t)).$$

Differentiate at $t = 0$:

$$(\nabla_X(fs))_p = \left. \frac{d}{dt} \right|_0 f(\gamma(t)) P_t^{-1} s(\gamma(t)) = X_p[f]s(p) + f(p)(\nabla_X s)_p.$$

This is exactly the scalar Leibniz rule with coefficient one. Hence the principal symbol is $\xi \otimes \text{id}_E$. $\square$

Corollary 11.2 - Exclusion of rescaled derivatives. Let $\Delta^{(c)} = c\nabla$ with $c \neq 1$. Then

$$\sigma_{\Delta^{(c)}}(\xi) = c\xi \otimes \text{id}_E,$$

so $\Delta^{(c)}$ fails the fixed first-jet normalization anchor J_D . If c is undeclared, it is imported normalization data. If c is declared, the operator grammar has changed. Only $c = 1$ remains in $\mathfrak{R}_D^1$.

This closes the proportional-symbol loophole. Identity principal symbol is not assumed. It is the derivative of finite carrier-linear transport together with the fixed base-scalar first jet.

12 Full connection identities

12.1 Additivity in the section argument

Theorem 12.1 - Section Additivity. For all sections s, t ,

$$\nabla_X(s + t) = \nabla_X s + \nabla_X t,$$

and $\nabla_X 0 = 0$.

Proof. By the Same-Scope Carrier-Morphism Preservation Theorem,

$$P_t^{-1}(s(\gamma(t)) + t(\gamma(t))) = P_t^{-1}s(\gamma(t)) + P_t^{-1}t(\gamma(t)).$$

Differentiating at 0 gives the result. The zero-section case is identical. $\square$

12.2 Linearity in the direction argument

Lemma 12.2 - Smooth Homogeneous First-Order Coefficient. Let V and W be finite-dimensional real vector spaces, and let $F : U \subseteq V \rightarrow W$ be smooth on a neighborhood U of 0. If

$$F(av) = aF(v)$$

whenever $v, av \in U$, then

$$F(v) = dF_0(v)$$

on the local domain. Hence F is linear there.

Proof. Fix v and define $\phi(t) = F(tv)$. Homogeneity gives $\phi(t) = tF(v)$ for all sufficiently small t . Differentiating at $t = 0$ gives

$$dF_0(v) = \phi'(0) = F(v).$$

Since dF_0 is linear, F agrees with a linear map on the local domain. $\square$

Theorem 12.3 - Direction Linearity. The canonical infinitesimalization is $C^\infty(M)$ -linear in the direction argument:

$$\nabla_{X+Y}s = \nabla_X s + \nabla_Y s, \quad \nabla_{fX}s = f\nabla_X s.$$

Proof. Work at p in a local frame. Proposition 10.3 gives a smooth first-order coefficient

$$A_p : T_p M \longrightarrow \text{End}(E_p),$$

where $A_p(v)$ is determined by the first-order expansion of transport along any path germ with initial velocity v . First-jet locality makes A_p a function of v alone. Reparameterization of path germs gives

$$A_p(av) = aA_p(v)$$

for all real a in the local tangent-germ domain; reversal supplies negative scalars. By Definition 8.1, admitted first jets are locally closed under rescaling and addition, so the argument applies on the tangent neighborhood required by the first-order target. Smoothness at 0 and Lemma 12.2 imply

$$A_p(v) = d(A_p)_0(v),$$

so A_p is linear. Hence $A_p(v + w) = A_p(v) + A_p(w)$ and $A_p(av) = aA_p(v)$. The raw derivative component $d_X s$ is already linear in X_p . Therefore $\nabla_X s = d_X s + A_p(X_p)s(p)$ is linear in X_p . Evaluating pointwise gives $\nabla_{X+Y}s = \nabla_X s + \nabla_Y s$ and $\nabla_{fX}s = f\nabla_X s$. $\square$

12.3 Connection identity package

Combining the identity principal-symbol theorem, the section-additivity theorem, and the direction-linearity theorem yields the full connection-identity package:

$$\nabla_X(s + t) = \nabla_X s + \nabla_X t,$$

$$\nabla_X(fs) = X[f]s + f\nabla_X s,$$

$$\nabla_{X+Y}s = \nabla_X s + \nabla_Y s,$$

$$\nabla_{fX}s = f\nabla_X s.$$

These are not inserted as axioms. They are consequences of smooth finite linear transport and the fixed first-jet scalar action.

13 Local connection normal form

Theorem 13.1 - Leibniz Difference Lemma. Let $E \rightarrow M$ be the standard finite linear carrier branch and let ∇ be the canonical infinitesimalization of an admissible smooth finite linear transport realization. In a local trivialization, let d_X be the raw component derivative. Define

$$A_X s = \nabla_X s - d_X s.$$

Then A is $C^\infty(M)$ -linear in both X and s , hence

$$A \in \Omega^1(\text{End } E), \quad \nabla = d + A$$

locally.

Proof. For the section argument,

$$A_X(fs) = \nabla_X(fs) - d_X(fs).$$

Using the identity principal-symbol theorem and the raw Leibniz rule,

$$A_X(fs) = (X[f]s + f\nabla_X s) - (X[f]s + fd_X s) = fA_X s.$$

Additivity follows from Theorem 12.1, while $C^\infty(M)$ -linearity in the direction argument follows from the direction-linearity theorem and the direction linearity

of d_X . Thus A is a one-form in the direction variable with values in fiber endomorphisms. $\square$

Corollary 13.2 - Jet sequence formulation. Equivalently, ∇ is a linear splitting of the first jet sequence

$$0 \longrightarrow T^*M \otimes E \longrightarrow J^1E \longrightarrow E \longrightarrow 0.$$

The identity principal symbol is precisely the statement that the splitting subtracts the base scalar first jet and leaves an endomorphism-valued correction.

14 Operator-family closure and adversarial examples

Same-scope operator exhaustion is applied only after the finite transport and infinitesimalization theorems have fixed the transport-realization candidate class.

Definition 14.1 - First-order transport-realization candidate classes. Let

$$\mathfrak{R}_D^1 = \{\Delta : \Delta \text{ infinitesimalizes admitted finite linear TC}_D\text{-transport}\}.$$

Equivalently, $\mathfrak{R}_D^1$ consists of smooth local first-jet infinitesimalizations of admissible finite linear standing-preserving transport realizations. Let

$$\mathfrak{O}_D^1 = \{\text{smooth local first-order operators on the presented carrier}\}.$$

In general, $\mathfrak{R}_D^1 \subsetneq \mathfrak{O}_D^1$. The theorem does not claim that every element of $\mathfrak{O}_D^1$ is a connection. It claims that every faithful same-scope realization of the finite transport role lies in $\mathfrak{R}_D^1$, and that every element of $\mathfrak{R}_D^1$ has the connection identity package.

Theorem 14.2 - Transport-Realization Operator Exhaustion. Let $\Delta \in \mathfrak{O}_D^1$. Then exactly one of the following applies:

1. $\Delta \in \mathfrak{R}_D^1$, hence Δ is the canonical connection induced by admissible finite linear transport;
2. Δ is a lawful redescription of such a realization;
3. Δ changes the first-jet normalization anchor;
4. Δ changes the linear carrier morphism graph;
5. Δ changes the differential order;
6. Δ changes the locality or continuation envelope; or
7. Δ introduces nonlocal, higher-jet, nonlinear, selector, boundary, source-side, or carrier-category enrichment.

Proof. If Δ is the canonical infinitesimalization of admissible finite linear TC_D -transport, then Sections 8-13 prove that Δ satisfies the connection identity package and has local form $d + A$. If it is presentation-equivalent to such an operator, it is a lawful redescription. Otherwise compare its operator graph with the frozen ledger $\mathfrak{S}_D^\nabla$. Failure of the scalar coefficient one changes J_D . Failure of fiberwise linearity changes the carrier morphism graph. Dependence on higher jets changes differential order. Dependence on remote, global, or path-history data changes the locality or continuation envelope. Nonlinear dependence, auxiliary selectors, boundary data, source data, or hidden normalization are operator-family enrichments. The same-scope operator closure theorem classifies all remaining purported extensions as bookkeeping, imported data, nonconservative family change, domain enlargement, or inadmissible repair. Therefore no nonconnection element of $\mathfrak{D}_D^1$ remains a faithful same-scope element of $\mathfrak{R}_D^1$. $\square$

The following examples are decisive regression tests.

Rescaled connection. $\Delta^{(c)} = c\nabla$, $c \neq 1$, fails the identity principal symbol. Its defect is imported normalization or changed first-jet anchor, not a new tensor field.

Nonlinear metric-section operator. On TM ,

$$\widetilde{\nabla}_X Y = \nabla_X Y + g(Y, Y)X$$

is smooth, local, and coordinate-covariant, but it is not the infinitesimalization of fiberwise-linear transport and is not additive in Y . It is nonlinear carrier-category enrichment using existing metric data.

Nonlinear direction operator.

$$\widehat{\nabla}_X Y = \nabla_X Y + g(X, X)Y$$

is smooth and local, but it changes the tangent-module morphism type and fails direction linearity. It is not a same-scope first-jet realization.

These operators are not dismissed because they disagree with the conclusion. They fail named fixed-ledger conditions: identity symbol, linear carrier morphism, or direction-module grammar.

15 Gauge, tangent, and tensor specializations

15.1 Gauge and associated carriers

Let a represented gauge carrier transform by

$$\psi' = g\psi.$$

A standing-preserving derivative must transform by

$$D'_\mu \psi' = g D_\mu \psi.$$

In a local representative,

$$D_\mu = \partial_\mu + B_\mu,$$

where B_μ is the represented endomorphism-valued coefficient induced by the connection representative on the associated carrier. Expanding

$$(\partial_\mu + B'_\mu)(g\psi) = g(\partial_\mu + B_\mu)\psi$$

gives

$$B'_\mu = g B_\mu g^{-1} - (\partial_\mu g) g^{-1}.$$

Sign, i , charge, and coupling conventions alter this display but not the dependency result. The term involving $\partial_\mu g$ is the derivative of redescription; the connection representative cancels it so that the derivative descends to standing-bearing carrier content.

15.2 Tangent carriers

For $E = TM$,

$$\nabla_\mu V^\nu = \partial_\mu V^\nu + \Gamma_{\mu\rho}^\nu V^\rho.$$

For a covector,

$$\nabla_\mu \omega_\nu = \partial_\mu \omega_\nu - \Gamma_{\mu\nu}^\rho \omega_\rho.$$

The sign pattern is fixed by contraction preservation:

$$\nabla_\lambda (\omega_\mu V^\mu) = \partial_\lambda (\omega_\mu V^\mu).$$

For a tensor $T_{\nu_1 \dots \nu_s}^{\mu_1 \dots \mu_r}$,

$$\nabla_\lambda T_{\nu_1 \dots \nu_s}^{\mu_1 \dots \mu_r} = \partial_\lambda T_{\nu_1 \dots \nu_s}^{\mu_1 \dots \mu_r} + \sum_{a=1}^r \Gamma_{\lambda\rho}^{\mu_a} T_{\nu_1 \dots \nu_s}^{\mu_1 \dots \rho \dots \mu_r} - \sum_{b=1}^s \Gamma_{\lambda\nu_b}^\rho T_{\nu_1 \dots \rho \dots \nu_s}^{\mu_1 \dots \mu_r}.$$

This is the tangent/tensor branch of the same local connection normal form.

16 Metricity as downstream comparison preservation

AASC does not begin with an independently complete metric and then infer governance. It begins with the governance roles required for determinate standing-bearing use. Metricity is downstream: it is a licensed faithful presentation of comparison when the governed comparison carrier is metric.

Definition 16.1 - Faithful metric branch. A faithful metric branch consists of (TM, g) inside an already governed comparison regime, with g serving as the standing-bearing comparison carrier for tangent content. Charts, frames, and coordinates are skin. Metric curvature and transport obstruction are tensorial content. Additional connection, torsion, nonmetricity, source, boundary, or higher-jet structures are present only when declared in the scope ledger.

Lemma 16.2 - Metric comparison preservation. If g is the admitted metric comparison carrier and no comparison-deformation occupant is declared, standing-preserving transport satisfies

$$X[g(Y, Z)] = g(\nabla_X Y, Z) + g(Y, \nabla_X Z),$$

equivalently

$$\nabla g = 0.$$

Proof. The metric supplies the standing-bearing comparison of tangent content. If transport changes that comparison without declaring a comparison-deformation occupant, then the transported object is no longer compared within the same metric role. The displayed equation is the coordinate-free expression of preserving metric comparison. $\square$

Metric-affine nonmetricity

$$Q_{\lambda\mu\nu} := -\nabla_\lambda g_{\mu\nu}$$

is not impossible. It is a declared post-metric comparison/connection tensor load. It is distinct from the AMetric boundary, where metric comparison is not yet licensed at all.

17 Natural Levi-Civita closure

The minimal metric branch must exclude primitive and derived contorsion. The phrase “no independent K ” is not sufficient, because a nonzero K might be built functorially from g using higher derivatives. The branch must therefore freeze the naturality and differential-order grammar.

Definition 17.1 - Minimal first-order natural metric branch. $\text{MinMetricNat}_1(M, g)$ holds when:

1. TM is the carrier;
2. g is the only metric-side standing-bearing selection datum;
3. the connection selection is local;
4. it is diffeomorphism-natural;
5. it depends on at most the first jet j^1g ;
6. it preserves metric comparison, $\nabla g = 0$; and
7. no orientation, volume form, preferred vector, foliation, boundary datum, source field, curvature datum, or nonlocal input is available.

This is the metric analogue of the minimal first-jet transport realization. Higher-derivative constructions are classified as higher-response metric-connection grammar.

Theorem 17.2 - Natural Difference-Tensor Vanishing. In the minimal first-order natural metric branch, any natural metric-compatible connection equals the Levi-Civita connection.

Proof. Let ∇ be a natural first-order metric-compatible connection. Since ∇^{LC} is also natural, first-order, and metric-compatible, their difference

$$K(X, Y) = \nabla_X Y - \nabla_X^{LC} Y$$

is a tensor

$$K \in \Gamma(T^*M \otimes T^*M \otimes TM).$$

Lower the upper index with g to obtain a natural covariant rank-three tensor $K_{\rho\mu\nu}$ depending only on j^1g . At any point p , choose normal coordinates for g :

$$g_{ij}(p) = \eta_{ij}, \quad \partial_k g_{ij}(p) = 0.$$

Because only j^1g is available, $K(p)$ can depend only on the model bilinear form η . Naturality requires $K(p)$ to be invariant under the full orthogonal group $O(p, q)$ of η . The element $-I \in O(p, q)$ acts on a covariant rank-three tensor by multiplication by -1 . Therefore

$$K(p) = -K(p),$$

so $K(p) = 0$. Since p was arbitrary, $K = 0$ and $\nabla = \nabla^{LC}$. $\square$

Corollary 17.3 - Torsion closure. In $\text{MinMetricNat}_1(M, g)$, torsion vanishes. Nonzero torsion requires a nonzero connection-difference tensor K , which Theorem 17.2 excludes in this branch.

Corollary 17.4 - Classification of derived contorsion. A candidate such as

$$V = dR, \quad K^\rho{}_{\mu\nu} = \delta^\rho_\mu V_\nu - g_{\mu\nu} V^\rho$$

is metric-compatible when added to ∇^{LC} , but it depends on curvature and dR , hence on higher jets of g . It belongs to a higher-response metric-connection grammar, not to $\text{MinMetricNat}_1(M, g)$.

Source of K	Disposition
$g, j^1 g$ only	Vanishes by Theorem 17.2.
Curvature or dR	Higher-jet/higher-response structure.
Orientation or volume form	Additional orientation anchor.
Preferred vector or foliation	Additional tensor or anchor.
Boundary or asymptotic data	Boundary/global enrichment.
Nonlocal functional of g	Continuation-envelope expansion.
Independently supplied torsion	Declared connection tensor load.

Thus Levi-Civita closure is not obtained by defining torsion away. It is the unique survivor of the minimal first-order natural metric grammar.

The coordinate representative is

$$\Gamma^\rho_{\mu\nu} = \frac{1}{2} g^{\rho\sigma} (\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu}).$$

18 Curvature and dynamics downstream

Once a connection representative exists, curvature is the loop residue of the licensed comparison structure. In the vector/gauge branch,

$$F = dA + A \wedge A, \quad [D_\mu, D_\nu]\psi = F_{\mu\nu}\psi.$$

In the tangent branch,

$$[\nabla_\mu, \nabla_\nu]V^\rho = R^\rho{}_{\sigma\mu\nu}V^\sigma$$

under the usual torsion-free convention. Thus the dependency order is

$$\text{TC}_D \Rightarrow \nabla = d + A \Rightarrow F \text{ or } R.$$

Curvature does not create the transport-comparison role. It is downstream tensor content produced by the already licensed comparison structure.

For Einstein dynamics, the metric branch supplies the Levi-Civita transport grammar. The Einstein-form theorem then concerns which metric first-response tensor can couple to a closed source ledger. That is a separate curvature-response exhaustion. For gauge-coupled Schrödinger or field dynamics, the replacement

$$\partial_\mu \mapsto D_\mu$$

is not a coupling trick; it is the representation-compatible derivative forced once the carrier occupies a nontrivial local redescription regime.

19 Main theorem

Theorem 19.1 - Kernel-to-Covariant-Differentiation Theorem. Let D be a determinate governed domain occupying $\text{CovDiffTarget}(D)$, with faithful standing descent, frozen covariant realization ledger $\mathfrak{S}_D^\nabla$, and the standard finite linear carrier branch $\text{StdFinLin}(D)$. Let the kernel-forced finite transport-comparison role TC_D admit an admissible smooth transport realization P satisfying identity, restriction, reversal, weak composition, smooth first-jet dependence, lawful-redescription equivariance, and preservation of the fixed carrier operations. Then the canonical infinitesimalization of P is a connection. Relative to every local trivialization,

$$\nabla = d + A, \quad A \in \Omega^1(\text{End } E).$$

No nonconnection first-order operator is a faithful same-scope realization. Such a candidate changes the jet anchor, carrier morphism graph, differential order, locality envelope, operator family, or declared scope.

In the minimal first-order natural metric branch,

$$\text{MinMetricNat}_1(M, g) + \nabla g = 0 \Rightarrow \nabla = \nabla^{LC}.$$

Proof. Determinate same-domain use instantiates the kernel by Imported Theorem 3.2. Target occupation forces admitted continuation, transportable carrier content, and smooth first-order refinement by Lemma 6.3. Finite route closure forces standing-preserving cross-local comparison through TC_D by Theorem 7.1. In the standard finite linear branch, an admissible smooth realization of TC_D is finite transport preserving identity, restriction, reversal, weak composition, redescription equivariance, first-jet dependence, and carrier operations. The Same-Scope Carrier-Morphism Preservation Theorem makes each transport map linear. Canonical infinitesimalization produces a well-defined local first-order operator. Differentiating finite carrier-linear transport gives section additivity and the identity principal symbol. Smooth first-jet dependence and reparameterization give direction linearity. The Leibniz Difference Lemma then

gives $A \in \Omega^1(\text{End } E)$ and $\nabla = d + A$ locally. Operator-family closure classifies every nonconnection first-order candidate as graph change, imported normalization, nonlinear carrier-category enrichment, higher-jet or nonlocal enrichment, bookkeeping, or explicit scope change. In the metric branch, metric comparison preservation gives $\nabla g = 0$, and the First-Order Natural Difference-Tensor Vanishing Theorem gives $\nabla = \nabla^{LC}$. $\square$

Corollary 19.2 - Local typed formulas. In the represented gauge branch,

$$D_\mu \psi = \partial_\mu \psi + B_\mu \psi.$$

In the tangent branch,

$$\nabla_\mu V^\nu = \partial_\mu V^\nu + \Gamma_{\mu\rho}^\nu V^\rho.$$

In the minimal first-order natural metric branch,

$$\nabla g = 0, \quad \nabla = \nabla^{LC}.$$

20 Richer-scope alternatives and nonclaims

The theorem does not assert that all physically meaningful theories must be torsion-free, metric-compatible, local, first-order, linear, or finite-rank. It classifies what is and is not a same-scope realization of the fixed target.

Structure	AASC status in this theorem
Raw partial derivative	Admissible only when it descends to standing-bearing content.
Smooth finite linear transport	Realization of TC_D in the standard branch.
Connection representative $c\nabla$, $c \neq 1$	Infinitesimalization of that transport. Imported normalization or changed first-jet anchor.
Nonlinear covariant operators	Nonlinear carrier-category or operator-graph enrichment.
Gauge connection	Associated-carrier branch of standing-preserving transport.
Affine connection	Tangent-carrier branch of standing-preserving transport.
Levi-Civita connection	Minimal first-order natural metric branch.

Structure	AASC status in this theorem
Torsion	Declared connection tensor load, unless excluded by the minimal branch.
Metric-affine nonmetricity	Declared post-metric comparison/connection load.
Higher-curvature or dR -built K	Higher-jet or higher-response metric-connection grammar.
Nonlocal connection selection	Continuation-envelope expansion.
Curvature	Downstream tensorial loop residue.

The theorem does not derive a particular manifold dimension, metric signature, gauge group, representation, coupling, global topology, boundary condition, field equation, or numerical connection. It derives the subordinate role and local type of the first-order correction when the relevant target and branch are occupied.

21 Refutation burden

A faithful theorem-level counterexample must identify one of the following failures.

1. The domain does not instantiate determinate same-domain use.
2. The imported kernel theorem fails on its own fixed-domain target.
3. The covariant-differentiation target is not occupied.
4. Faithful standing descent is absent or misidentified.
5. The finite route exhaustion has an unclassified same-scope survivor.
6. A proposed smooth realization preserves finite linear transport but fails the connection identities.
7. The identity principal-symbol theorem is invalid.
8. A nonconnection operator belongs to $\mathfrak{R}_D^1$ without changing carrier graph, first-jet anchor, differential order, locality, or scope.
9. A nonzero natural first-order metric-compatible difference tensor K exists using only g and j^1g .
10. The theorem reports beyond its declared standard finite linear or minimal metric branch.

A downstream physical theory with extra fields, torsion, nonmetricity, nonlocality, higher-curvature terms, or nonlinear carrier morphisms is not a counterexample unless it preserves the frozen ledger. Otherwise it is a declared richer-scope theory.

22 Conclusion

Covariant differentiation is not primitive in AASC, and it is not a familiar formula reinterpreted after the fact. It is the smooth first-jet realization of a kernel-forced standing-preserving transport-comparison role.

The dependency order is

$$\text{DetUse}(D) \Rightarrow K_D \Rightarrow \mathcal{Q}_D,$$

$$\text{CovDiffTarget}(D) \Rightarrow \text{TC}_D,$$

and, in the standard finite linear carrier branch with admissible smooth transport realization,

$$\text{TC}_D \Rightarrow P \Rightarrow \sigma_{\nabla}(\xi) = \xi \otimes \text{id}_E \Rightarrow \nabla = d + A.$$

The identity principal symbol, additivity, direction linearity, and local endomorphism-valued one-form correction are derived from finite carrier-linear transport. They are not inserted as connection axioms. Nonconnection operators fail as same-scope realizations by named criteria: changed normalization, changed carrier morphism graph, nonlinear carrier-category enrichment, higher-jet enrichment, nonlocal enrichment, or explicit scope change.

In the metric branch, governance remains upstream and metricity is downstream licensed comparison. In the minimal first-order natural metric grammar, metric comparison preservation gives $\nabla g = 0$ and the natural difference-tensor theorem excludes every nonzero first-order natural contorsion. The survivor is Levi-Civita. Torsionful, metric-affine, higher-jet, and nonlocal structures remain legitimate declared richer-scope branches.

Thus the endpoint is preserved in its strongest defensible form: once the covariant-differentiation target, standard finite linear carrier, and smooth standing-preserving transport realization are fixed, connection type is forced, locally $\nabla = d + A$; and once the minimal first-order natural metric branch is fixed, the metric specialization is ∇^{LC} .

Appendix A. Finite transport antecedent

This appendix records the finite transport theorem in the form used by the main text, so the covariant-differentiation result is auditable without external access to the companion transport manuscript.

A.1 Fixed finite transport problem

Let $\mathcal{C}_D$ be the continuation site. Objects are local contexts $U, V, \dots$ carrying presentations of the standing-bearing carrier. A continuation arrow is a typed admitted morphism

$$\kappa_{UV} : U \rightsquigarrow V.$$

Let $\mathcal{C}_D^{\text{cmp}}$ be the class of same-domain cross-local comparison candidates over admitted arrows. A candidate belongs to this class only if it preserves the frozen finite comparison ledger:

1. fixed target;
2. carrier class;
3. standing quotient or descent interface;
4. admissibility standard;
5. continuation relation;
6. fixed comparison role;
7. transformation ledger; and
8. fixation and exclusion status.

A candidate changing any of these coordinates is not a finite same-scope alternative. It is a declared scope enrichment or a different target.

A.2 Candidate dependency signatures

For each $C_{UV} \in \mathcal{C}_D^{\text{cmp}}$, record the dependency signature

$$\Sigma(C_{UV}) = (R, B, S, W, M, E, Rep, Sk, Fail),$$

where:

- R marks representative dependence not factoring through descent;
- B marks transition bookkeeping without transported carrier content;
- S marks a selector, preferred representative, privileged chart, frame, trivialization, or gauge;
- W marks absence of an admitted continuation witness or scope-bounded transport witness;
- M marks quotient-preserving mediated continuation;
- E marks enrichment by undeclared carrier, tensor, source, boundary, operator, comparison, or selection data;
- Rep marks same-act repair or retrospective certification;
- Sk marks skin or presentation-only change; and
- $Fail$ marks fail-closed nonstanding.

The classifier is a coverage device, not a priority algorithm. A candidate may receive multiple terminal dispositions.

A.3 Mediator conditions

A mediator Θ_κ over $\kappa : U \rightsquigarrow V$ is a finite transport-comparison witness only if it satisfies:

1. quotient-domain condition: it is defined on descended classes or factors through the descent interface;
2. continuation condition: it is attached to the admitted continuation arrow κ ;
3. carrier-role preservation: standing-bearing carrier role is preserved from source to target;
4. no-selector condition: no privileged representative is chosen outside the allowed transition ledger;
5. restriction compatibility: shrinking U, V restricts the same witness class;
6. weak composition: admitted composable arrows compose up to the declared finite comparison equivalence;
7. identity compatibility: identity arrows act as standing-preserving identity comparisons; and
8. no-extra-load condition: no independent carrier, tensor, source, boundary, selector, operator, or comparison structure is introduced.

Write

$$\text{TC}_D(\kappa) = [\Theta_\kappa]$$

for the class of such witnesses.

A.4 Coverage theorem

Finite Coverage Theorem. For every $C_{UV} \in \mathcal{C}_D^{\text{cmp}}$, either C_{UV} receives at least one terminal disposition among $R, B, S, W, E, Rep, Sk, Fail$, or it factors through a mediator satisfying the conditions of A.3.

Proof. If the output of C_{UV} varies under replacement of representatives with the same descended standing content, then R holds. If it records only overlap or transition bookkeeping and supplies no transported carrier content, then B holds. If it resolves comparison by privileging a representative, then S holds. If it asserts cross-context comparison without an admitted continuation witness, then W holds. If it succeeds only by adding undeclared structure, then E holds. If it relies on later certification of a failed comparison, then Rep holds. If it changes no standing-bearing content, then Sk holds. If it yields no admitted standing-bearing comparison, then $Fail$ holds. The only remaining case is a quotient-facing, continuation-attached, carrier-preserving, selector-free, restriction-compatible, weakly compositional, identity-respecting, no-extra-load mediator. That is exactly the M case. $\square$

A.5 No seventh route

No Seventh Finite Route Corollary. If a cross-local comparison candidate avoids representative sensitivity, bookkeeping-only status, selector import, witnesslessness, enrichment, same-act repair, skin, and fail-closed status, then it factors through TC_D .

Proof. The finite coverage theorem leaves only the mediator case. By A.3, every such mediator is a representative of TC_D . $\square$

A.6 Fixation closure

Fixation Closure Lemma.

$$\neg \text{Fix}_D \Rightarrow \neg \text{StableExclusion}_D \Rightarrow \neg \text{RouteClosure}_D.$$

Proof. Without Fixation, an excluded representative comparison can be reopened as if it had not failed, a selector can be retroactively treated as admitted, a later witness can certify a pathless assignment within the same act, and an enriched carrier can be reidentified as the original carrier. Then the terminal dispositions in the coverage theorem are not stable. Fixation is the role preventing same-act repair and preserving the endpoint classification. $\square$

A.7 Finite transport theorem used in the main text

Finite Transport-Comparison Theorem. If D instantiates the kernel, faithful standing descent, admitted continuation, local presentation freedom, and the fixed finite comparison ledger, then every same-scope standing-preserving cross-local comparison factors through TC_D .

Proof. Apply the finite coverage theorem. All nonmediated active cases are terminal failures or scope changes. Fixation closes their terminal status. The only same-scope standing-preserving survivor is quotient-preserving mediated continuation, which is TC_D . $\square$

This antecedent is pre-differential. It contains no A_μ , $\Gamma_{\mu\nu}^\rho$, curvature, holonomy, or covariant derivative. Those appear only after a smooth finite linear transport realization is admitted and infinitesimalized.

The full finite-transport proof is developed in *Connection as Standing-Preserving Transport Geometry* [5]. The present appendix reproduces only the portion needed for the covariant-differentiation theorem. The current paper’s new contribution is the smooth first-jet infinitesimalization and connection-representative classification.

Appendix B. Adversarial regression suite

Candidate	Threat	Blocking theorem or branch condition	Disposition
$\Delta = c\nabla, c \neq 1$	A rescaled local covariant operator might appear to satisfy the same first-order requirements.	Identity Principal-Symbol Theorem; fixed first-jet normalization anchor J_D .	First-jet normalization change; not a faithful same-scope infinitesimalization.
$\nabla_X Y + g(Y, Y)X$	A nonlinear operator can be built from already declared metric data without adding a new field.	Same-Scope Carrier-Morphism Preservation Theorem; Transport-Realization Operator Exhaustion.	Nonlinear carrier-category enrichment; not an infinitesimalization of finite fiberwise-linear transport.
$\nabla_X Y + g(X, X)Y$	Direction dependence may be nonlinear while remaining smooth and coordinate-covariant.	Direction-Linearity Theorem; Operator-Family Closure.	Direction-module graph change; not a first-jet tangent-linear transport infinitesimalization.
Derived dR -contorsion K	Contorsion can be metric-derived.	Natural Difference-Tensor Vanishing; MinMetricNat ₁ .	Higher-jet/higher-response metric connection load.
Nonlocal $K[g]$	A nonlocal connection selection may preserve covariance while using global metric data.	Realization-ledger locality and first-jet condition.	Continuation-envelope expansion; not a same-scope first-jet realization.
Orientation-dependent $K[g, \varepsilon]$	Orientation or density data may produce covariant-looking connection corrections.	First-Order Natural Metric Difference-Tensor Vanishing Theorem; fixed metric-branch anchor ledger.	Additional orientation anchor; richer metric branch.

Candidate	Threat	Blocking theorem or branch condition	Disposition
Preferred-foliation connection	A foliation or preferred vector could select a compatible correction.	Fixed local first-order natural metric branch; no additional vector/foliation anchor.	Additional vector/foliation anchor; scope enrichment.
Boundary-selected connection	Boundary data could select a correction.	Fixed local first-order natural metric branch.	Boundary/global enrichment.
Nonlinear finite transport	Transport may be functorial while not preserving the linear carrier graph.	Same-Scope Carrier-Morphism Preservation Theorem.	Changed carrier morphism category; declared nonlinear branch if retained.
Path rescale	Path rescaling might change the scalar coefficient.	Identity Principal-Symbol Theorem; fixed first-jet normalization.	Changed first-jet normalization anchor.

The derived dR -contorsion row covers, for example, $K^\rho_{\mu\nu} = \delta^\rho_\mu (dR)_\nu - g_{\mu\nu} (dR)^\rho$. Each case is blocked by a named theorem, branch condition, or frozen ledger coordinate. None is rejected merely because it differs from the desired conclusion.

Appendix C. Conventional normalization

The ordinary mathematical normalizations used above are standard. A vector-bundle connection is equivalently a first-order linear differential operator with identity principal symbol, or a splitting of the first jet sequence. A smooth parallel-transport functor with the usual locality and smoothness conditions gives a connection by infinitesimalization, and a connection gives such transport by solving the parallel-transport equation. The Levi-Civita connection is the unique torsion-free metric connection, and more strongly the unique first-order natural metric-compatible connection in the minimal metric grammar used here.

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